DETECTING WHAT KNOWS YOU'RE LOOKING

Adversarial anomaly detection under correlated backgrounds

A math-first deck with plain-language read-along explanations

THE QUESTION

How do you detect a hidden actor that knows how you are trying to detect it?

v1.4.2 protected-covariance benchmark
Independent, reproducible research

Important: the speculative origin story is not evidence that any concealed population exists.

How to read this deck

GUIDE

Every technical slide has a formal layer and a plain-language layer.

FORMAL LAYER

- Probability models, Bayes factors, KL divergence, combinatorics
- Stackelberg games, linear programs, covariance geometry
- Mahalanobis tests, noncentral χ^2 power, Monte Carlo invariants
- Exact equations are shown whenever they drive a result

HOME-FOLKS LAYER

- What does this equation actually mean?
- Why did the previous idea fail?
- What should a detector do differently?
- What does the result not prove?

PLAIN ENGLISH

The deck is intentionally redundant: the same idea is expressed as an equation, an algorithm, and an ordinary-language intuition.

Source: verified project versions v1.0–v1.4.2; synthetic models unless noted.

01 PART I

From a speculative premise to a general detection problem

The original thought experiment supplied an adversary. The mathematics quickly stopped depending on the motivating story.

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formal mathematics on the left • plain-language translation on the right

Separate existence, behavior, and motive

LOGIC

A first logical cleanup prevented us from smuggling conclusions into the premises.

SYMBOLIC DISTINCTIONS

$$E \not\Rightarrow G, \qquad I \not\Rightarrow G, \qquad M \not\Rightarrow I$$

E = existence / presence
M = monitoring
I = intervention
G = governance

Evidence for one layer must not be reused as evidence for another without an explicit model.

PLAIN ENGLISH

Even if a hidden actor existed, that would not tell us whether it rules, intervenes, monitors, helps, harms, or cares. Those are separate questions.

CAUTION

This deck studies the detection problem. It does not assert that the motivating hidden population exists.

Source: verified project versions v1.0–v1.4.2; synthetic models unless noted.

The abstract game

MODEL

An observer watches noisy channels; an adaptive hider chooses a footprint and may contaminate controls.

OBSERVER

- chooses sensors / controls
- chooses a statistical test
- may randomize the test
- estimates the background

WORLD / BACKGROUND

$$X \sim \mathcal{P}_0(\Sigma)$$

Ordinary variation has a joint structure, not just independent noise.

HIDER

- chooses signal size s
- chooses footprint direction u
- may reach r controls
- may adapt to the observer

PLAIN ENGLISH

The core research object is not “a weird event.” It is a strategic interaction between a detector, an ordinary background, and an actor trying to blend into that background.

Source: verified project versions v1.0–v1.4.2; synthetic models unless noted.

Baseline: signal buried in a noisy count

PROBABILITY

The earliest model made the signal-to-background problem explicit.

FORMAL MODEL

$$Y \mid N, d, B \sim \text{Poisson}(B + Nd), \qquad \mathbb{E}[Y] = B + Nd$$

READ IT LIKE THIS

Y is the number of candidate events we observe. B is ordinary background. N is the hidden population and d is per-agent detectability. Only the product Nd appears in the count.

√1000 ≈ 32
Poisson SD when B = 1000

100
example background SD

20
example extra signal

Source: verified project versions v1.0–v1.4.2; synthetic models unless noted.

Evidence is a likelihood ratio, not a feeling

BAYES

Bayesian model comparison forced us to integrate over uncertainty instead of choosing convenient parameter values after the fact.

FORMAL MODEL

$$\frac{P(H_i \mid E, S)}{P(H_0 \mid E, S)} = \frac{P(E \mid H_i, S)}{P(E \mid H_0, S)} \frac{P(H_i \mid S)}{P(H_0 \mid S)}$$

READ IT LIKE THIS

Posterior odds = Bayes factor × prior odds. If two hypotheses predict the observation equally well, the data do not move you. Flexible models must average over all their parameter settings, including the bad ones.

Example from the early count model

Fixed signal / known background: BF 1.220 → uncertain background: 1.037 → marginalized N,d,B: 0.979

Source: verified project versions v1.0–v1.4.2; synthetic models unless noted.

02

PART II

The adversary chooses its detectability

Once the hidden actor can adapt, detectability is no longer merely a property of the sensor. It becomes a strategic choice constrained by operating cost.

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formal mathematics on the left • plain-language translation on the right

Stealth as an optimization problem

KL DIVERGENCE

We borrowed the logic of information-theoretic steganography: distinguishability is a property of whole distributions.

FORMAL MODEL

$$d^* = \arg\min_d \left[D_{\mathrm{KL}}(P_{1,d} \| P_0) + \kappa \log\left(\frac{d_{\max}}{d}\right) \right]$$

READ IT LIKE THIS

The hider dislikes being statistically distinguishable, but concealment also costs something. κ tells us how expensive it is to reduce the detectable footprint.

Interpretation The detector pushes d downward; operating cost pushes d upward. The equilibrium lives where those pressures balance.

CAUTION The cost function is a modeling assumption. Changing it changes the equilibrium detectability—an important lesson from our sensitivity sweeps.

Source: verified project versions v1.0–v1.4.2; synthetic models unless noted.

A closed form exposed what the cost function was doing

CLOSED FORM

For small signals in a Gaussian-like background, KL behaves quadratically.

$$D_{\mathrm{KL}} \approx \frac{s^2}{2V}, \qquad J(s) = \frac{s^2}{2V} + \kappa \log\left(\frac{100}{s}\right)$$

$$\frac{dJ}{ds} = 0 \quad \Rightarrow \quad s^* = \sqrt{\kappa V}, \qquad D_{\mathrm{KL}}^* = \frac{\kappa}{2}$$

PLAIN ENGLISH

If the background gets 11× noisier, the optimal hidden footprint grows by √11 ≈ 3.32. But under this specific logarithmic cost, the hider adjusts until its optimal statistical detectability is unchanged.

PLAIN ENGLISH

This was a key turning point: “better background calibration improves detection” was not universally true. Under this cost form, calibration mainly suppresses how much activity the hider can afford.

Source: verified project versions v1.0–v1.4.2; synthetic models unless noted.

Why one count channel is not identifiable

IDENTIFIABILITY

Repeated measurements can estimate a sum perfectly without telling us how the sum is divided.

FORMAL MODEL

$$Y_i \sim \text{Poisson}(B + s) \quad \Rightarrow \quad \text{data identify } B + s, \text{ not } B \text{ and } s \text{ separately}$$

READ IT LIKE THIS

More observations of the same channel reduce random error, but they do not tell us which part of the mean is background and which part is the hidden signal. We need a contrast—a channel where the signal is absent or constrained.

ON/OFF idea

Observe an ON channel and one or more OFF controls. The controls are not “free truth”; they can be heterogeneous, contaminated, or strategically chosen against you.

Source: verified project versions v1.0–v1.4.2; synthetic models unless noted.

Coverage: how many controls can the hider reach?

COMBINATORICS

Randomization helps only when the hider cannot contaminate every possible control.

FORMAL MODEL

P(all measured controls contaminated) = C(r,k) / C(m,k), for k ≤ r

P(all measured controls contaminated) = 0, for k > r

where C(a,b) = “a choose b” = number of ways to choose b items from a.

READ IT LIKE THIS

There are m candidate controls. The observer secretly samples k of them. The hider can reach r. If k > r, at least one measured control must be clean.

Example: m=8, r=4

k=1 → 50% all contaminated • k=2 → 21.4% • k=4 → 1.43% • k=5 → 0%

Source: verified project versions v1.0–v1.4.2; synthetic models unless noted.

The control set can be attacked through its variance

ROBUST STATISTICS

A balanced +δ/−δ spread attack leaves the control mean alone while inflating the estimated scale.

NON-ROBUST SCALE

$$\widehat{\sigma}_{\mathrm{SD}} = \sqrt{\frac{1}{k-1}\sum_i (x_i - \bar{x})^2}$$

A hider can increase this by pushing a few controls apart.

ROBUST SCALE

$$\hat{\sigma}_{\mathrm{MAD}} \propto \mathrm{median}_i \left| x_i - \mathrm{median}(x) \right|$$

MAD tolerates substantial contamination before breaking, but it sacrifices efficiency on clean data.

THE LESSON

The observer is not choosing one “correct” statistic. It is trading clean-data efficiency against adversarial robustness.

PLAIN ENGLISH

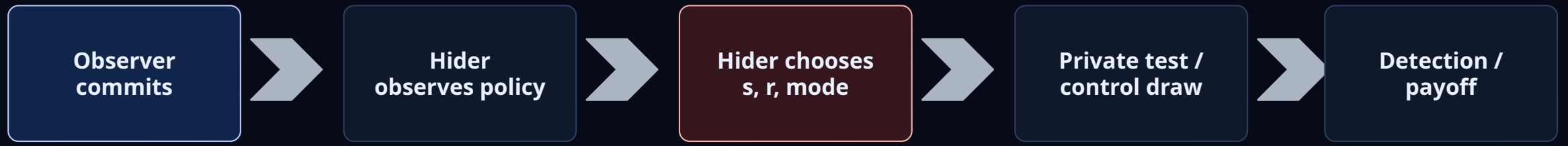
This SD-vs-MAD tradeoff later reappeared at the test level: directional tests are efficient against plain activity; omnibus tests are harder to contaminate.

Source: verified project versions v1.0–v1.4.2; synthetic models unless noted.

The Stackelberg game

GAME THEORY

The observer commits first; the hider best-responds after seeing the commitment, but not necessarily its private random realization.



$$U_H(a \mid d) = \text{activity benefit} - \text{detection penalty} - \text{operating cost}$$

$$d^* = \arg\min_d \; L_D\left(d, \mathrm{BR}_H(d)\right)$$

PLAIN ENGLISH

Stackelberg is the right structure whenever the detector can publish a policy but keep the actual randomized choice secret. The hider must choose one response against a distribution of possible tests.

Source: verified project versions v1.0–v1.4.2; synthetic models unless noted.

Willingness to pay isolates the value of contamination

WTP

Regime counts were misleading because they depended on arbitrary contamination-cost constants.

FORMAL MODEL

$$\mathrm{WTP}(d,\tau) = \max_{a \in \mathcal{A}_C} U_H(a \mid d, c=0) - \max_{a \in \mathcal{A}_0} U_H(a \mid d)$$

READ IT LIKE THIS

Ask how much extra utility the hider gains if contamination is free. That gain is the maximum price it would rationally pay for contamination capability.

Why it helped

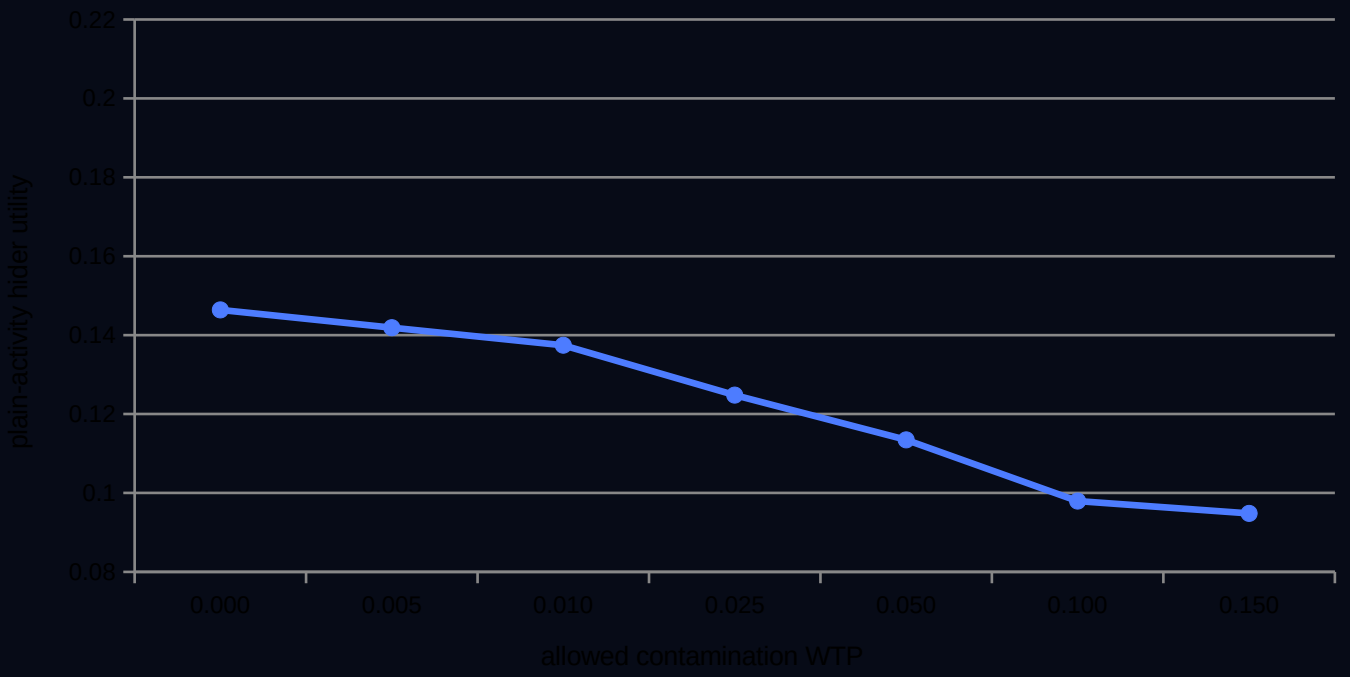
WTP compares detector architectures without assuming a particular contamination price. Grid-ceiling cases are explicitly marked as lower bounds.

Source: verified project versions v1.0–v1.4.2; synthetic models unless noted.

Robustness–efficiency frontier

FRONTIER

A detector can suppress plain activity, resist contamination, or save measurement effort—but usually not maximize all three at once.



WHAT THE FRONTIER MEANS

- Lower plain-activity utility is better for the observer.
- Lower WTP is also better.
- Randomized test mixtures moved the frontier inward.
- Expected number of controls E[k] is a third axis.

PLAIN ENGLISH

Randomization is useful because the hider must choose one attack before knowing which test will actually run. That denies it a perfectly tailored response.

Source: verified project versions v1.0–v1.4.2; synthetic models unless noted.

03

PART III

Covariance is the geometry of “ordinary”

Noise is not a scalar. What matters is how ordinary activity moves jointly across channels—and whether the hidden footprint points along a noisy or quiet direction.

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formal mathematics on the left • plain-language translation on the right

COVARIANCE

A covariance family that separates common-mode and clustered noise

This is the v1.4 protected-background model.

FORMAL MODEL

$$\Sigma = vI + \tau^2 \left[\rho \mathbf{1}\mathbf{1}^T + (1-\rho)\,\mathrm{blockdiag}(J_c)\right]$$

READ IT LIKE THIS

v is independent channel noise. τ^2 is structured variation. ρ moves that variation into a global common mode. c is cluster size. At ρ=0,c=1 we recover the independent scalar-heterogeneity model.

Key identity **Common-mode noise often cancels in contrasts; cluster-specific noise creates genuinely weak directions that whitening cannot erase.**

Source: verified project versions v1.0–v1.4.2; synthetic models unless noted.

Contrast, whiten, then test

MAHALANOBIS

Known covariance lets the observer weight each direction by how noisy it really is.

$$z = Cx, \qquad \Omega = C\Sigma C^T Q \mid H_0 \sim \chi^2_k$$

$$Q = z^T \Omega^{-1} z Q \mid H_1 \sim \chi^2_k(\lambda)$$

PLAIN ENGLISH

Subtract the controls from the ON channel, then scale each resulting direction by how much ordinary variation that direction normally has. A 10-unit deviation in a quiet direction is more suspicious than 10 units in a naturally noisy one.

PLAIN ENGLISH

Whitening removes covariance as a nuisance only in the sense of correct weighting. It cannot turn a genuinely noisy physical direction into a quiet one.

Source: verified project versions v1.0–v1.4.2; synthetic models unless noted.

Noncentrality λ is the signal-to-noise geometry

POWER

Protected-regime power is analytic once the footprint direction is specified.

FORMAL MODEL

$$\lambda = \mu_z^T \Omega^{-1} \mu_z, \qquad \text{Power} = P\left(\chi_k^2(\lambda) > q_{1-\alpha}\right)$$

READ IT LIKE THIS

λ is squared Mahalanobis distance: how far the signal moves us in units of the background covariance. Larger λ means easier detection.

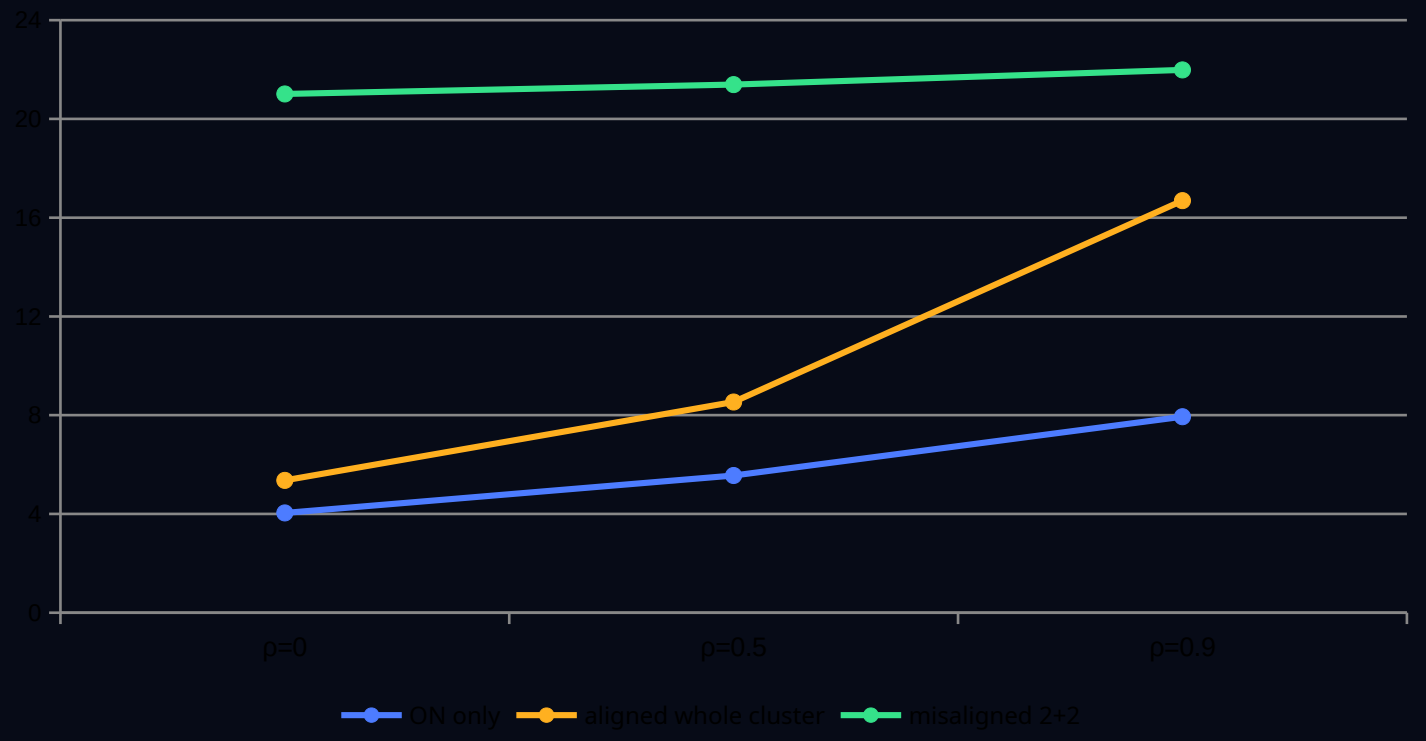
5.36	21.01	16.69
aligned c=4, ρ=0	misaligned 2+2	aligned c=4, ρ=0.9

Source: verified project versions v1.0–v1.4.2; synthetic models unless noted.

Cluster alignment can help or hurt the hider

GEOMETRY

At fixed marginal variance, common-mode correlation can improve detection while cluster-aligned attacks remain intrinsically weak.



READ THE CHART

- At ρ=0, a whole-cluster attack hides in a high-variance mode.
- Splitting the same attack across clusters is much easier to detect.
- As ρ rises, more noise becomes common mode and cancels in contrasts.

Verified values from protected_attack_geometry_v1_4_2.csv.

The compact result: concealment capacity

RESULT

Normalize the physical footprint so rescaling u cannot game the definition.

$$C^*(\Sigma, \mathcal{U}) = \max_{u \in \mathcal{U},\ \|u\|=1} \frac{1}{\sqrt{u^T \Sigma^{-1} u}}$$

Special case: every unit direction is feasible

$$C^* = \sqrt{\lambda_{\max}(\Sigma)}$$

PLAIN ENGLISH

The best hiding place is the physically feasible direction where ordinary behavior has the most unexplained variance after all calibrated cross-channel relationships are taken into account.

PLAIN ENGLISH

A very noisy city can still offer little concealment if its noise is well modeled. A quieter system can offer more cover if the relevant direction is poorly characterized.

Source: verified project versions v1.0–v1.4.2; synthetic models unless noted.

04

PART IV

Verification is part of the mathematics

Several attractive conclusions failed under reproduction, alternative seeds, finer grids, or different linear-algebra backends. The verifier is not an appendix—it is a result.

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formal mathematics on the left • plain-language translation on the right

The computational pipeline

ALGORITHMS

Exact analysis where possible; Monte Carlo only where it buys something.

1. signals s, reach r, test methods, covariance family
2. exact Gaussian / noncentral χ^2 when Σ is protected
3. activity benefit − detection penalty − operating cost
4. hider chooses the highest-utility action
5. mixed commitment under WTP and E[k] constraints
6. simulation, regression cells, checksums, portability

PLAIN ENGLISH

The research loop was: propose → formalize → code → reproduce → attack assumptions → version. Several “results” were demoted to illustrations when sensitivity analysis showed they depended on arbitrary utilities.

Source: verified project versions v1.0–v1.4.2; synthetic models unless noted.

Protected-benchmark invariants

VERIFICATION

The current v1.4.2 benchmark has five independent checks.

Invariant	Status	Observed / basis
Model reduction ρ=0,c=1	PASS	exact matrix identity
Common-mode contrast identity	PASS	$C(\Sigma+\gamma 1 1^{T})C^{T} = C\Sigma C^{T}$
Simulated null size	PASS	0.009995 at α=0.01
Analytic vs simulated power	PASS	max 1.84 Monte Carlo SE
675-cell v1.3.3 regression	PASS	mean z=0.793; SD=0.959

Repeated eigenvalues made seeded SVD sampling non-portable across LAPACK builds. Cholesky fixed the actual reproducibility problem.

PORTABLE RNG

Cholesky, not SVD

Verified from protected_invariants_v1_4_2.csv and protected_simulation_checks_v1_4_2.csv.

Next: finite-T covariance estimation

V1.4 NEXT

Once Σ is estimated rather than known, the reference distribution changes.

KNOWN MEAN, ESTIMATED COVARIANCE

$$\frac{T-k}{k(T-1)}Q \sim F_{k,T-k}$$

ESTIMATED MEAN + ESTIMATED COVARIANCE

$$\frac{T(T-k)}{k(T-1)(T+1)}Q \sim F_{k,T-k}$$

Requires T > k. Reusing χ^2 thresholds at small T would inflate false alarms.

WHAT WE WANT TO MEASURE

$$\Delta_T(w) = U^*_{\text{plain}}(w;\hat{\Sigma}_T) - U^*_{\text{plain}}(w;\Sigma)$$

How much extra vulnerability comes purely from having to learn the background instead of knowing it?

PLAIN ENGLISH

This is the first quantity that can answer a practical design question: “How many clean calibration periods do we need before covariance uncertainty stops dominating the detector?”

Source: verified project versions v1.0–v1.4.2; synthetic models unless noted.

A live-demo algorithm for AI InTheRock

DEMO

Use a real multichannel dataset, inject a controlled hidden footprint, and let the audience watch the detector succeed or fail.



PLAIN ENGLISH

The visual payoff: show that a signal can be obvious in one plot and nearly invisible after the hider rotates it into a noisy covariance direction—then show how whitening or randomized tests recover some of the lost information.

Target datasets: multichannel telemetry with meaningful covariance and enough clean calibration history.

What this framework says—and what it does not

BOUNDARIES

The project is methodological, not evidentiary.

IT DOES SAY

- A passive detector can be non-identifiable against a perfectly distribution-matched actor.
- Limited reach, physical constraints, and private randomization can restore identifiability.
- Correlated backgrounds create weak and strong directions.
- Calibration uncertainty can be quantified relative to a protected benchmark.

IT DOES NOT SAY

- That a cryptoterrestrial or extraterrestrial population exists.
- That any unexplained anomaly is evidence of an intelligent actor.
- That “more sensors” helps unless covariance is modeled.
- That one cost function, estimator, or test is universally optimal.

PLAIN ENGLISH

The honest endpoint is a testable design problem: define the background, define feasible footprints, define the actor’s reach, and ask what observations actually distinguish the models.

Source: verified project versions v1.0–v1.4.2; synthetic models unless noted.

THE MATH IN FIVE LINES

1. **Evidence is a likelihood ratio, not an anomaly count.**
2. **A strategic actor chooses a footprint, not just a location.**
3. **Controls only help when their joint covariance is modeled.**
4. **Randomization denies the hider a perfectly tailored response.**
5. **Concealment is geometry: $1 / \sqrt{(u^{T}\Sigma^{-1}u)}$.**

OUR ASK

A real multichannel dataset.

A domain expert who can break the model.

A chance to show what modest compute + disciplined AI-assisted reasoning can accomplish.

adversarial-anomaly-detection • reproducible Python • current milestone: v1.4.2 protected covariance

05 APPENDIX

Earlier models and equations

These exploratory models helped clarify the problem, even when later versions replaced or narrowed them.

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formal mathematics on the left • plain-language translation on the right

Appendix A — secrecy as a time-varying hazard

HISTORICAL MODEL

A simple leakage model quantified how population, disclosure rate, and verification interact.

FORMAL MODEL

$$P(L = 0) = \exp\left[-\int_0^T N(t)\,\lambda(t)\,p(t)\,dt\right]$$

READ IT LIKE THIS

N(t) is the resident population, λ(t) is per-agent leakage hazard, and p(t) is the chance a leak becomes independently verifiable. Long time alone is not enough; detectability through time matters.

CAUTION

This model was exploratory. Human secrecy-rate estimates do not transfer automatically to a hypothetical nonhuman population.

Source: verified project versions v1.0–v1.4.2; synthetic models unless noted.

Appendix B — participation vs biological detectability

HISTORICAL MODEL

Embedding more deeply in society can increase the chance of entering biological sampling pipelines.

FORMAL MODEL

$$\mu_{genomic} = Nfsg, \qquad P(0\ \text{detections}) \approx e^{-\mu_{genomic}}$$

READ IT LIKE THIS

N is population; f is deep-integration fraction; s is sampling probability; g is probability that the pipeline would actually surface the biological difference instead of filtering it out.

PLAIN ENGLISH

The important correction was that g includes the analysis pipeline. A sample can be biologically unusual yet be discarded as contamination or QC failure.

Source: verified project versions v1.0–v1.4.2; synthetic models unless noted.

Appendix C — threshold intervention model

HISTORICAL MODEL

We briefly modeled intervention probability as a smooth threshold function of risk.

FORMAL MODEL

$$q_H(r) = \frac{1}{1 + e^{-k_H(r - r_H)}}$$

READ IT LIKE THIS

r is a model-specific risk score. r_H is the intervention threshold. k_H controls how sharply behavior changes around that threshold. Different motives imply different thresholds.

CAUTION This equation was a placeholder for model comparison, not an empirical law. Without a defensible risk definition and objective function, it predicts nothing.

Source: verified project versions v1.0–v1.4.2; synthetic models unless noted.

Appendix D — cost-form sensitivity

SENSITIVITY

The logarithmic concealment cost was special; power-law costs make calibration affect optimized detectability.

LOG COST

$$c(s) \propto \log(1/s) \quad \Rightarrow \quad D_{\mathrm{KL}}^{*} = \kappa/2 \,(\text{independent of } V)$$

POWER-LAW COST

$$c(s) \propto s^{-\alpha} \quad \Rightarrow \quad D_{\mathrm{KL}}^{*} \propto V^{-\alpha/(2+\alpha)}$$

PLAIN ENGLISH

The statistics do not determine the equilibrium by themselves. The hider’s operating-cost curve can be the dominant assumption. That is why later versions swept utilities and reported cost-free WTP.

PLAIN ENGLISH

Sensitivity analysis converted several headline claims from “result” to “illustration.” That is a success, not a failure.

Source: verified project versions v1.0–v1.4.2; synthetic models unless noted.

Appendix E — directional, omnibus, and combined tests

TEST FAMILY

The observer’s action set contains tests with different robustness profiles.

DIRECTIONAL

$$z = \frac{X_{\mathrm{ON}} - \bar{X}_{\mathrm{OFF}}}{\sqrt{\mathrm{Var}(X_{\mathrm{ON}} - \bar{X}_{\mathrm{OFF}})}}$$

Strong against plain ON elevation; can be fooled when controls move with the signal.

OMNIBUS

$$Q = z^T \Omega^{-1} z$$

Tests whether the channels remain mutually consistent; partial contamination often becomes signal.

COMBINED

$$\text{reject if } Z > z_{1-\alpha/2} \vee Q > q_{1-\alpha/2}$$

Bonferroni union: compromise between efficiency and robustness. Later mixed commitments randomize across tests instead of trusting one.

PLAIN ENGLISH

There is no universal winner. The relevant object is the frontier over plain-activity vulnerability, contamination WTP, and measurement effort.

Source: verified project versions v1.0–v1.4.2; synthetic models unless noted.

Appendix F — mixed-policy frontier as a linear program

LP

The observer’s randomized commitment is solved exactly on the finite strategy grid.

$$\min_{x} \max_{a \in \mathcal{A}_0} x^T U(a)$$

$$\text{s.t.} \quad \max_{a \in \mathcal{A}_C} x^T U(a) - \max_{a \in \mathcal{A}_0} x^T U(a) \leq w$$

$$1^T x = 1, \quad x \geq 0, \quad \mathbb{E}_x[k] \leq K$$

Implementation trick Enumerate which uncontaminated action attains the inner maximum; each region becomes an ordinary linear program. The best feasible region gives the exact mixed frontier on the finite grid.

WHY MIXING HELPS

The hider sees the distribution x but not the realized test. It must choose one attack that works reasonably well against the whole mixture.

Example at τ=0, WTP cap 0, k≤8: 69% combined + 31% omnibus plain-activity utility ≈ 0.146

Exact finite-grid LP, verified in v1.3.5.